

* Deformation theory, Homological Algebra & Mirror Symmetry:

- Deformation theory Also known as the theory of moduli

* Moduli: are related to the def. of schemes stacks & their $\mathbb{C}$ analytic analogues

Classical Deformation theory:

• $M: \mathbb{C}\text{-Mfd}$

$\Lambda_{\mathbb{C}}^k M = \bigoplus_{p+q=k} \Lambda^{p,q} M$: Decomp. of the set of the $\mathbb{C}$ -valued k -forms

$\Omega^{p,q}(M)$: set of all smooth sections of $\Lambda^{p,q} M$

* $\mathbb{C}$ -structure is characterized by the Dolbeault differential

$$\bar{\partial}: \Omega^{p,q}(M) \rightarrow \Omega^{p,q+1}(M)$$

$R :=$ Commutative ring w/ unit

Def: A differential graded Alg (DGA) over R is a triple $(A^*, \cdot, d)$ with

(i) for each $k \in \mathbb{Z}_{\geq 0}$, A^k is an R -module
 $\deg(a) = k$ if $a \in A^k$

(ii) $\cdot: A^k \otimes A^l \rightarrow A^{k+l}$ is an R -module homo
Namely, $(a \cdot b) \cdot c = a \cdot (b \cdot c)$

(iii) $d: A^k \rightarrow A^{k+1}$ is an R -module homo

$$\text{s.t. } d \circ d = 0$$

$$(iv) \quad d(a \cdot b) = d(a) \cdot b + (-1)^{\deg(a)} a \cdot d(b)$$

We may take $(\oplus \Omega^{p,q}(M), \wedge, \bar{\partial})$, K : degree

$$O\mathcal{D} \left(\sum_{p,q} \Omega^{p,q}(M), \wedge, \bar{\partial} \right)$$

• Let $\pi_E: E \rightarrow M$ be a $\mathbb{C}$ -vector Bundle
 $\rightarrow \Omega^{p,q}(M; E) \equiv \Gamma(M, \wedge^{p,q}(M; \wedge^{p,q}(M \otimes E)))$

$$\wedge: \Omega^{p,q}(M) \otimes \Omega^{p',q'}(M; E) \rightarrow \Omega^{p+p',q+q'}(M; E)$$

$$\alpha \wedge \beta$$

Def: A holomorphic structure on E is a seq. of operators $\bar{\partial}_E: \Omega^{p,q}(M; E) \rightarrow \Omega^{p,q+1}(M; E)$
 $p, q \in \{1, \dots, n\}$ s.t.

(i) $\bar{\partial}_E \circ \bar{\partial}_E = 0$

(ii) $\bar{\partial}_E(u \wedge \alpha) = \bar{\partial} u \wedge \alpha + (-1)^{p+q} u \wedge \bar{\partial}_E(\alpha)$
 for $u \in \Omega^{p,q}(M)$, $\alpha \in \Omega^*(M; E)$

In other words the holomorphic structure on E is a structure on $\Omega^*(M; E)$ of a left graded diff. graded module over $(\Omega(M), \wedge, \bar{\partial})$

